IEM KOLKATA

Linear Algebra
Matrices & Determinants

Q1) If $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}$, then $\det(A^{-1})$ is

(a) 0.25 (b) 0.50 (c) 0.75 (d) 1.25

Q2) Let $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{bmatrix}$.

Let det(A) and det(B) denote the determinants of the matrices A and B, respectively. Which one of the options given below is TRUE?

(a) det(A) = det(B)

(b) det(B) = - det(A)

(c) det(A) = 0

(d) det(AB) = det(A) + det(B)

Q3) The matrix $A = \begin{bmatrix} a & 0 & 3 & 7 \\ 2 & 5 & 1 & 3 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & b \end{bmatrix}$ has det(A) = 100 and trace(A) = 14. The value of $\mid a - b \mid$ is

(a) 1 (b) 2 (c) 3 (d) 4

Q4) If any two columns of a determinant $= \begin{vmatrix} 4 & 7 & 8 \\ 3 & 1 & 5 \\ 9 & 6 & 2 \end{vmatrix}$ are interchanged, which one of the following statements regarding the value of the determinant is CORRECT?

(A) Absolute value remains unchanged but sign will change
(B) Both absolute value and sign will change
(C) Absolute value will change but sign will not change
(D) Both absolute value and sign will remain unchanged

Q5) Consider the following two statements with respect to the matrices Am x n, Bn x m, Cn x n, Dn x n.
Statement 1 : tr(AB) = tr(BA)
Statement 2 : tr(CD) = tr(DC)
where tr( ) represents the trace of a matrix. Which one of the following holds?
(A) Statement 1 is correct and Statement 2 is wrong
(B) Statement 1 is wrong and Statement 2 is correct
(C) Both Statement 1 and Statement 2 are correct
(D) Both Statement 1 and Statement 2 are wrong

Q6) Let $P = \begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}$ and $Q = P^3 - 2P^2 - 4P + 13I_2$, where I_2 denotes the identity matrix of order 2. Then the determinant of Q is equal to _____________ (answer in integer).

Q7) Let $ABC = I$, then $B^{-1} =?$

Q8) Let A, B, C, D be $n \times n$ matrices, each with non-zero determinant. If $ABCD = I$, then $B^{-1} =?$

Q9) X, Y are singular matrices $\ni XY = Y$ and $YX = X$. Then $X^2 + Y^2 =?$

Q10) If $\mid A^{3\times 3} \mid = 3$, then $\mid (2A)^{-1} \mid =?$

Q11) If $\mid A^{3\times 3} \mid = 4$, then $adj(adj(A)) =?$

Q12) Consider the system of simultaneous equations:

$$x + 2y + z = 6$$

$$2x + y + 2z = 6$$

$$x + y + z = 5$$

The system has:
(a) unique solution
(b) infinite no of solutions
(c) no solution
(d) exactly two solutions

Q13) Consider the system of simultaneous equations:

$$\begin{bmatrix} 2 & 1 & -4 \\ 4 & 3 & -12 \\ 1 & 2 & -8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \alpha \\ 5 \\ 7 \end{bmatrix}$$

Notice that the second and the third columns of the coefficient matrix are linearly dependent. For how many values of α, does this system of equations have infinitely many solutions?
(a) 0 (b) 1 (c) 2 (d) infinitely many